

Drug Utilisation Studies in the OMOP Common Data Model

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Table of contents

Preface	1
Is this book for me?	1
How is the book organised?	1
Citation	2
License	2
Code	2
Packages version	2
1 Introduction	3
1.1 DrugUtilisation Studies	3
2 Proposed structure for the book	4

Preface

Is this book for me?

This book is part of the collection of **Tidy R programming in OMOP** see accompanying books:

- (1) Tidy R programming with the OMOP common data model
- (2) The omopverse developer book

We've written this book for anyone interested to conduct DrugUtilisation studies using the OMOP Common Data Model and working with databases using a tidyverse style approach. That is, human centered, consistent, composable, and inclusive (see <https://design.tidyverse.org/unifying.html> for more details on these principles).

New to R? We recommend you compliment the book with R for data science

New to databases? We recommend you take a look at some web tutorials on SQL, such as SQLBolt or SQLZoo

New to the OMOP CDM? We'd recommend you pare this book with The Book of OHDSI

How is the book organised?

TO ADD

Citation

Package	Version	Link
DrugUtilisation	1.0.4	
omock	0.5.0	

Citation

TO ADD

License

Code

The source code for the book can be found at this Github repository

Packages version

This book is rendered using the following version of packages:

```
Finding R package dependencies ... Done!
```

Note we only included the packages called explicitly in the book.

1 Introduction

1.1 DrugUtilisation Studies

- What is a DrugUtilisation Study in general

2 Proposed structure for the book

- Preface
 - Is this book for me
 - * OMOP
 - * R
 - * Tidyverse
 - * Tidy R in OMOP CDM collection
 - * Set up your environment
 - About the authors
- Introduction
 - Drug Utilisation studies
 - Drug data in the OMOP CDM
 - Synthetic data used
- Drug Users identification
 - Create drug codelists
 - * Stratify and subset codelists
 - * Import and export codelists
 - Diagnostics
 - * DrugExposureDiagnostics
 - * PhenotypeR
 - * Dose calculation
 - Identify users
 - * Identify new users

2 Proposed structure for the book

- Population level DrugUtilisation
 - Denominator cohort
 - * General Population
 - * Target Population
 - * External Denominator
 - Incidence
 - * Standardised Incidence Rate
 - * Incidence Rate Ratio
 - * Standardised Incidence Rate Ratio
 - Prevalence
 - * Point Prevalence
 - * Period Prevalence
- Patient level DrugUtilisation
 - New users characterisation
 - * Characterisation
 - * Large Scale Characteristics (Data Driven Characterisation)
 - Indication
 - * Indication in the data base
 - * Phenotype Driven indication
 - * Data Driven indication
 - Drug use
 - * How to calculate daily dose?
 - * Summarise drug use
 - Treatments
 - * Alternative treatments
 - * Comedications
 - Discontinuation
 - * As survival
 - * Proportion of patients covered

2 Proposed structure for the book

- Drug switching
 - * Drug restart and drug switching
 - * Treatment pathways
 - * Multi-state
- Dose escalation
 - * Dose summary
 - * Dose pathways
 - * Dose states
- DrugUtilisation study
 - Final considerations
 - Examples of DrugUtilisation Studies
 - Other studies

```
library(DrugUtilisation)
library(omock)
```